Notification System

Admin manages all notifications from one screen on the website.

They do not need to open WhatsApp, Postmark, or push provider websites to create templates.

Everything is done from the admin panel.

1. Trigger

A trigger is anything that should cause a notification to go out.

It is not only login or logout. It can be any event or condition on the website, for example:

Trigger example	When it fires
Login	User signs in on the website
Logout	User signs out
Not logged in for 1 day	User has not visited the website for 24 hours
Not logged in for 1 week	User has not visited for 7 days
Password reset	User asks to reset password
Order placed	User completes a purchase

One trigger = one row in the admin table.

You can add as many triggers as the website needs.

2. Channels (where the message goes)

Each trigger can send on three channels:

Channel	What the user gets	Service used (in background)
WhatsApp	Message on WhatsApp	WhatsApp Cloud API
Email	Email in inbox	Postmark (or another free transactional email API — see Section 4)
Web Push	Pop-up in the browser	Web Push

Note: We use Web Push only (browser).

When a trigger fires, the website can send on one, two, or all three channels — each with its own template.

Example — trigger: Login

User logs in on website

- WhatsApp: "Welcome back!"
- Email: "You logged in successfully"
- Web Push: "Welcome back!" (browser notification)

Example — trigger: Not logged in for 1 week

User away for 7 days

- WhatsApp: "We miss you, come back!"
- Email: "It's been a week..."
- Web Push: "Come visit us again"

3. Admin screen (one table)

Trigger	WhatsApp	Email	Web Push
Login	template	template	template
Logout	template	template	template
Not logged in 1 day	template	template	template
Not logged in 1 week	template	template	template

- Row = one trigger
- Column = one channel
- Each cell = one template for that trigger on that channel

Admin can from each cell:

- Create template
- Edit template
- Turn on / off
- Test send

4. Sandbox setup (for candidates — no real API keys)

Candidates will not get real WhatsApp Cloud API or production Postmark keys.

Use sandbox / free accounts only.

WhatsApp — Sandbox

1. Create a free account on Meta for Developers (developers.facebook.com).
2. Create an app → add WhatsApp product.
3. Use the test phone number and temporary token from API Setup.
4. Add your own phone number to the test recipient list (only those numbers can receive test messages).
5. Put in .env:

```
WHATSAPP_ACCESS_TOKEN=your_test_token
PHONE_NUMBER_ID=your_test_phone_number_id
```

Token expires often — generate a new one from Meta when tests fail.

Email — Postmark (free)

6. Sign up at Postmark (postmarkapp.com).
7. Create a developer server — free tier gives 200 emails free (enough for practice).
8. Verify your email as sender.
9. Put in .env:

```
POSTMARKAPP_TOKEN=your_server_token
POSTMARK_FROM_EMAIL=your_verified_email@example.com
```

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Other free email API options

Postmark's free tier is limited (roughly 100 emails/month on current plans), which can feel tight once you're testing several trigger templates repeatedly. Candidates may use any of the following free alternatives instead — the integration pattern (verified sender + API token in .env) is the same for all of them:

Provider	Free tier	Notes
Postmark	~100 emails/month	Named in this doc; best transactional deliverability; strict separation from marketing email
Brevo	300 emails/day (~9,000/month)	Most generous free tier for this assignment's testing volume; simple REST API
Resend	3,000 emails/month	Modern developer experience, good fit for React/Next.js frontends
Mailgun	100–300 emails/day	Developer-focused, built-in email validation
Amazon SES	62,000/month (first 12 months, from EC2)	Cheapest at scale, but more setup; only free if sending from EC2

If you switch providers, update the .env variable names to match (e.g. Brevo uses BREVO_API_KEY and a verified sender email) and note which provider you used in your README.

Web Push — Free tier

10. Sign up at OneSignal (onesignal.com) (free).
11. Create a website app → enable Web Push only (skip Android / iOS).
12. Subscribe in your browser to get a subscription ID for test sends.
13. Put in .env:

```
ONESIGNAL_APP_ID=your_app_id
ONESIGNAL_REST_API_KEY=your_api_key
```

Do not commit .env or share tokens.

5. Practice tasks

Task A — Pick any trigger (e.g. Login)

14. Open Notification Settings in admin.
15. Find the Login row (or any trigger row).
16. Set up all three channels:

Channel	Steps
WhatsApp	Create Template → save → Sync until approved → Test to your phone
Email	Create Template → subject + body → save → Test to your email

Channel	Steps
Web Push	Create Template → title + body → turn off iOS and Android → save → Test in browser

✅ Done when you get test message on phone, email, and browser.

Task B — Second trigger (e.g. Logout or Not logged in 1 week)

17. Pick another row (Logout, or inactive-user trigger).
18. Do the same for WhatsApp, Email, Web Push.
19. Use different message text than Task A.

✅ Done when second row has three working templates.

Task C — Edit and toggle

20. Edit one template → change text → test again.
21. Turn off one channel toggle → turn on again.

✅ Done when you understand edit + on/off.

Task D — Explain (no notes)

Answer in simple words:

22. What is a trigger? Give 3 examples (not only login).
23. What are the three channels?
24. Why create templates in admin panel instead of Postmark / WhatsApp site?
25. What is Web Push?

6. FAQ

Q: Where do I create templates?

A: Only in admin Notification Settings. The system talks to WhatsApp Cloud API, Postmark, and Web Push for you.

Q: Does every trigger send all three channels?

A: Only if all three cells have templates, toggles are on, and the website code fires that trigger.

Q: Can I add a new trigger?

A: Yes. New row → fill WhatsApp, Email, Web Push. Same steps every time.

Q: Real WhatsApp / Postmark account?

A: No. Use sandbox and free tier only for practice.

Q: Mobile app push?

A: No. Web Push in browser only.

7. Assignment — what you must build

Time: 2 days

AI tools: Allowed — use any AI tool you want (ChatGPT, Cursor, Copilot, etc.)

You must build the full notification system on frontend and backend.

Part	Tech	Host on
Backend	Python + Django	Render
Frontend	Your choice (React, Next.js, etc.)	Vercel

Backend (Django on Render)

- Admin APIs for triggers and templates
- Connect to WhatsApp Cloud API (sandbox), Postmark or another free email API (free), Web Push (free)
- Fire notifications when a trigger happens on the website
- Store triggers, templates, on/off toggles, variable mappings

Frontend (on Vercel)

- Website where users can log in, log out, etc. (triggers happen here)
- Admin panel — one table: rows = triggers, columns = WhatsApp / Email / Web Push
- Create, edit, test send, and turn channels on/off from the admin panel only
- Web Push subscribe in the browser

Minimum triggers to support

At least 2 triggers, for example:

- Login
- Logout

Or any others from section 1 (e.g. not logged in for 1 day / 1 week).

Each trigger must work on all 3 channels when toggles are on.

Sandbox only

Use sandbox / free accounts (section 4). No production WhatsApp or Postmark keys.

What to submit

26. GitHub repo (or two repos — backend + frontend)
27. Live backend URL (Render)
28. Live frontend URL (Vercel)
29. Short README: how to log in as admin, which triggers you built, env vars needed

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30. End-to-end walkthrough video with voice explanation — screen-recorded, narrated demo covering: logging in as admin, creating/editing a template, firing each trigger, and receiving the notification on all 3 channels (WhatsApp, Email, Web Push). Share as an unlisted/public link (e.g. Loom, Google Drive, YouTube unlisted).

8. Checklist

- ☐ Sandbox WhatsApp set up + test phone added
- ☐ Postmark (or other free email API) account set up + sender verified
- ☐ Web Push free account + browser subscribed
- ☐ Django backend built and deployed on Render
- ☐ Frontend + admin panel built and deployed on Vercel

- ☐ At least 2 triggers with all 3 channels working
- ☐ Task A — one trigger, all 3 channels tested
- ☐ Task B — second trigger, all 3 channels tested
- ☐ Task C — edited template + used toggle
- ☐ Task D — can explain triggers and channels
- ☐ GitHub + live URLs submitted
- ☐ Walkthrough video (voice-narrated, end-to-end) recorded and link submitted